

MemeCap: Cross-Modal Retrieval, Metaphor Classification, and Sentiment Analysis

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Project & Dataset Overview

- ▶ **Objective:** Study how vision and language models interpret complex visual metaphors and internet irony.
- ▶ **Dataset (MEMECAP):** 6,384 memes with titles, literal/metaphorical captions, and entity mappings.
 - ▶ **Train Split:** 5,823 memes — **Test Split:** 559 memes
- ▶ **Multi-Task Project Scope:**
 - ▶ **Task 2.1:** Cross-Modal Caption Retrieval
 - ▶ **Task 2.2:** Literal vs. Metaphorical Caption Classification
 - ▶ **Task 2.3:** Multiclass Sentiment Analysis (7-Class)

Task 2.1: Cross-Modal Retrieval Framework

- ▶ **Goal:** Retrieve the metaphorical meme caption given a meme image (Type 1) or an image + title (Type 2).
- ▶ **Evaluation Metrics:** Recall@1, Recall@5, and Mean Reciprocal Rank (MRR).
- ▶ **Zero-Shot Baselines:** Deployed OpenCLIP (ViT-L/14) and SigLIP2.
 - ▶ *Type 2 Late Fusion:* $q = \text{normalize}(\alpha \cdot e_{\text{image}} + (1 - \alpha) \cdot e_{\text{title}})$.
 - ▶ *Reranking:* Deployed BLIP Image-Text Matching (ITM) as an auxiliary ranker on zero-shot candidates to boost top precision.
- ▶ **End-to-End Fine-Tuning:**
 - ▶ **From Scratch (CNN-BiGRU):** 4-block Residual CNN + Bidirectional GRU trained via Symmetric Contrastive Loss.
 - ▶ **LoRA Fine-tuning:** Low-Rank Adaptation (Rank=16, Alpha=32) targeting attention weights of pretrained OpenCLIP.

Task 2.1.c: Custom Architecture Details

Encoder Architectures Evaluated

- ▶ **Image Encoder:** Implemented a 4-block Residual CNN (ResNet-style) with Adaptive Average Pooling. Chosen over standard CNNs to prevent vanishing gradients during early epochs.
- ▶ **Text Encoder:** Utilized a Bidirectional GRU (BiGRU) with both Mean and Max pooling to efficiently capture sequential meme context and strong keyword signals.

Fusion Strategies for Type 2 (Image + Title)

- ▶ **Attempt 1 (Linear Fusion):** Simple element-wise addition/averaging of image and title vectors. Failed to capture the complex, non-linear relationships between a Reddit title and a meme.
- ▶ **Final Choice (MLP Fusion):** Concatenated CNN and BiGRU embeddings, passing them through a FeatureFusion layer equipped with **GELU activations** and **Layer Normalization** to project down to a 256-dimensional joint space.

Training Dynamics & Challenges

Task 2.1: Retrieval Results Summary

Model Configuration	Input Type	R@1 (%)	R@5 (%)	MRR (%)
OpenCLIP Zero-Shot	Type 1	60.29	74.78	67.51
OpenCLIP Zero-Shot	Type 2	56.71	71.38	63.81
OpenCLIP + BLIP Reranker	Type 2	68.16	78.89	73.16
SigLIP2 Zero-Shot	Type 1	54.74	70.84	62.54
Custom CNN-BiGRU (From Scratch)	Type 2	0.54	1.61	1.82
OpenCLIP LoRA Fine-tuned	Type 1	68.16	79.96	73.66
OpenCLIP LoRA Fine-tuned	Type 2	66.91	82.47	74.02

- **Takeaway:** LoRA parameter-efficient fine-tuning significantly outperformed both zero-shot baselines and custom models trained from scratch.
- Data scarcity ($\approx 5,800$ samples) makes learning metric embeddings from scratch extremely difficult, while LoRA successfully shifts pre-trained weights.

Task 2.2: Literal vs. Metaphorical Classification

The Keyword Bias Challenge (Zero-Shot)

- ▶ Relying on raw CLIP cosine similarity yields a high F1 (0.800) but an extremely poor **ROC-AUC of 0.241**.
- ▶ **Why?** Metaphorical and literal captions share exact visual entities (nouns), fooling a simple linear similarity comparison.

Custom Fusion Architecture Solution

- ▶ **Features:** Extracted frozen 768-dimensional visual and textual OpenCLIP embeddings, normalized and concatenated.
- ▶ **Classifier:** Late Fusion MLP with LayerNorm, GELU, and Dropout. Trained with Binary Cross-Entropy.
- ▶ **Outcome:** The dedicated MLP head successfully resolves keyword bias by learning non-linear boundary relationships.

Task 2.2: Ablation Study & Results

Architecture Variation	Accuracy	F1-Score	ROC-AUC
Zero-Shot (1 – Sim)	0.667	0.800	0.241
MLP Base (BatchNorm + ReLU)	0.987	0.991	0.999
Ablation: Focal Loss	0.983	0.988	0.999
Ablation: Advanced Fusion	0.983	0.989	0.999
Final Selected Model (LayerNorm)	0.989	0.993	0.999

► Ablation Findings:

- LayerNorm proved significantly more effective than BatchNorm for handling dense Transformer-based representations.
- Transitioning to GELU activations and tuned dropout stabilized hidden layers, bringing performance to **98.9% Accuracy** and **99.3% F1**.

Task 2.3.a: Multiclass Sentiment Annotation

- ▶ **Goal:** Annotate the entire dataset with 7 granular emotion categories to establish target labels for training downstream classifiers.
- ▶ **Hurdle 1: Neutral Bias (Text-Only)**
 - ▶ Pure text models (RoBERTa) mapped $\approx 55\%$ of memes to "Neutral," as they are blind to the humorous visual template.
- ▶ **Hurdle 2: Visual Literalism (Base VLMs)**
 - ▶ LLaVA-1.5 mapped $> 54\%$ of memes to "Sadness" because it interpreted visual crying templates literally. Even with advanced *Few-Shot* and *Sarcasm* prompts, it failed to grasp irony.
- ▶ **The Solution: Context-Aware VLM (Qwen-VL-Chat)**
 - ▶ We utilized Qwen-VL-Chat, which natively possesses a much stronger understanding of internet culture.
 - ▶ **Outcome:** Without relying on heavy few-shot examples, it successfully captured internet sarcasm, shifting the distribution to a balanced spread dominated by **Joy (43.22%)**.

Task 2.3.a: Class Imbalance & Label Distribution

Emotion	DistilRoBERTa	Twitter-RoBERTa	LLaVA Base	LLaVA Few-Shot	LLaVA Sarcasm	Qwen-VL (Final)
Neutral	54.31%	54.56%	6.78%	19.48%	0.09%	20.68%
Joy / Positive	5.39%	7.54%	4.40%	13.29%	3.37%	43.22%
Sadness	7.36%	-	54.84%	35.26%	86.60%	3.45%
Disgust	19.26%	-	20.29%	19.62%	1.38%	4.31%
Anger / Negative	6.66%	37.90%*	2.10%	2.52%	4.95%	8.63%
Surprise	3.12%	-	6.36%	6.75%	2.05%	16.08%
Fear	3.90%	-	5.22%	3.09%	1.55%	3.64%

- **Key Takeaway:** Text-only models default to Neutral due to lack of vision. Base LLaVA suffers from severe visual literalism. Qwen-VL-Chat's sarcasm-aware few-shot prompting is the only setup that resolves internet irony.

Task 2.3.a: Qualitative Labeling Analysis

- **Overcoming Irony:** Examples of how Qwen-VL-Chat successfully interpreted contextual irony rather than relying on literal visual descriptions.



Label: Anger
Correct



Label: Joy
Correct

Me back in Highschool



Me currently in my Master's



Label: Joy
Wrong

Task 2.3.b & 2.3.c: Sentiment Classifiers

Training Setup (Unimodal vs. Multimodal Fusion)

- ▶ **Task 2.3.b Unimodal Baselines:** Independent MLPs trained on L2-normalized 768-dimensional frozen CLIP visual or textual embeddings.
- ▶ **Task 2.3.c Custom Late Fusion MLP:** Concatenated joint input (1536-dim), trained with high regularization (**0.5 Dropout, 0.1 Label Smoothing, 1e-2 Weight Decay**) to combat early overfitting.

Model	Accuracy	Macro F1
Image-Only MLP	0.4043	0.2725
Text-Only MLP	0.5224	0.4251
Late Fusion MLP	0.5134	0.4211

Key Finding: Modality Dominance

- ▶ Text is the primary emotional "anchor".
- ▶ Recycled visual templates introduce statistical noise (modal interference) when combined, making Late Fusion slightly worse than Text-Only.

Key Project Challenges & Strategic Solutions

Challenge 1: Metric Learning on Data Scarcity (Task 2.1.c)

- ▶ Training a dual-encoder from scratch on $\sim 5,800$ samples yielded poor retrieval.
- ▶ **Solution/Proposal:** Auxiliary classification losses (using metaphorical or sentiment labels) to guide representations via Multi-Task Learning.

Challenge 2: Overcoming Visual Literalism (Task 2.3.a)

- ▶ Base VLMs interpreted sarcastic templates literally, failing to grasp irony even when we applied advanced few-shot sarcasm prompting.
- ▶ **Solution:** Transitioning to **Qwen-VL-Chat** for its native, zero-shot understanding of internet culture—though it introduces a slight "Joy Imbalance" due to the inherently humorous nature of memes.

Summary & Next Steps

► Overall Achievements:

- Successfully applied LoRA to OpenCLIP for efficient cross-modal retrieval.
- Solved metaphorical classification (0.99 F1) using Late Fusion.
- Addressed VLM visual literalism to generate high-quality sentiment labels.

► What is Left? (Architectural Next Steps):

► Task 2.1 Custom Architecture Revamp:

- *Multi-Task Learning*: Train encoders with a joint objective, adding metaphorical classification as an auxiliary loss to learn richer shared representations.
- *Hard Negative Mining*: Inject semi-hard negatives in symmetric contrastive loss to force the CNN and BiGRU to learn fine-grained template-text alignments.
- *Cross-Attention Early Fusion*: Employ transformer-based cross-attention layers instead of late fusion, allowing visual attention to be guided dynamically by the text tokens.

► Task 2.3 Label Generation Refinement:

- *VLM Ensemble Strategy*: Combine LLaVA (which over-indexes on Sadness) and Qwen (which over-indexes on Joy) via a consensus or weighted voting mechanism to achieve a fully balanced and robust sentiment distribution.

Thank You!

Questions & Answers